

Breaking AES at 50 km: Long-Range Optical Side-Channel Attack over Fiber

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Abstract *Passive optical networks rely on AES-128 for traffic confidentiality. We demonstrate that AES-induced power fluctuations on microcontrollers can modulate a co-powered laser and propagate through 50 km of fiber. This enables key recovery using 1500 optical traces, making this the first long-range optical side-channel attack. ©2026 The Authors*

Introduction

The massive deployment of Passive Optical Networks (PON) provides very high-speed internet access to millions of subscribers. The PON downstream traffic is broadcast to all users, creating an opportunity for attacks. To prevent malicious users from accessing broadcast data intended for other users, the Gigabit-capable Passive Optical Networks (G-PON) [1] and 10-Gigabit-capable Symmetric Passive Optical Networks (XGS-PON) [2] standards mandate the use of the Advanced Encryption Standard (AES) [3]. While AES is cryptographically robust, hardware or software implementations are known to leak physical information through side-channels, i.e., the unintentional information leakage that occurs as a side effect of how computer hardware/software is used.

Side-Channel Analysis (SCA) can exploit power consumption variations or electromagnetic emanations that correlate with the data processed by a cryptographic device. Recently, optical emissions have emerged as a particularly subtle source of unintended emissions. It has been shown that the light emitted by a power LED, captured with an off-the-shelf camera, can reveal side-channel signatures exploitable for key recovery [4].

SCA can also leverage radio-frequency emissions induced by cryptographic operations, transforming a device's communication interface into an unintended wireless leakage channel with a range of approximately 10 meters [5, 6].

In this paper, we study SCA approaches in the optical communication domain. We investigate

whether AES-induced electrical power supply fluctuations can propagate through a pigtailed DFB laser over long fiber distances. We build an experimental setup where an Arduino executing AES-128 and a DFB laser share the same power supply. We measure optical traces at the receiver side after the propagation of the optical signal in a 50km single mode fiber. We demonstrate that we can recover the full AES-128 key using Correlation Power Analysis (CPA) [7] with 1500 traces. To the best of our knowledge, this is the first optical side-channel attack at long distance, demonstrating that information leakage can propagate far beyond the immediate physical module environment.

Experimental Setup

The experimental setup is depicted in Fig. 1. An Arduino Uno (ATmega328P, 16 MHz clock) is used to execute a standard-compliant implementation of AES-128 [3]. The board is powered by a linear laboratory power supply at DC 5V. The laser is gated on the first SubBytes operation. This non-linear substitution step of AES produces the strongest data-dependent power variations and is therefore the most exploitable operation for side-channel analysis. Enabling the laser only during this window introduces a sharp rising edge in the optical signal, which serves as a trigger for the oscilloscope and ensures that all traces are temporally aligned on the same cryptographic operation. For each acquisition, the secret key is kept fixed while the plaintext is randomized.

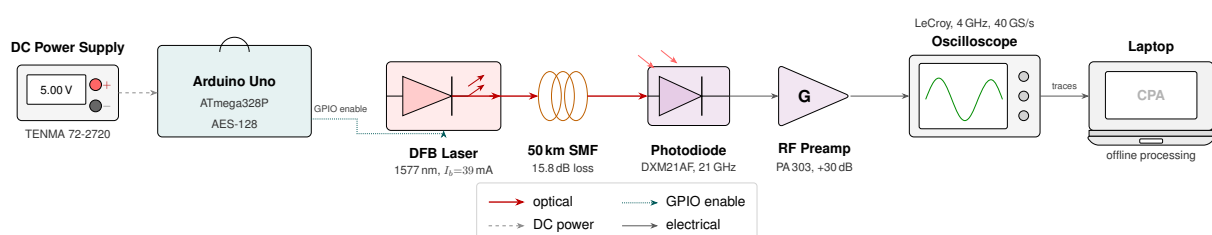


Fig. 1: Experimental setup.

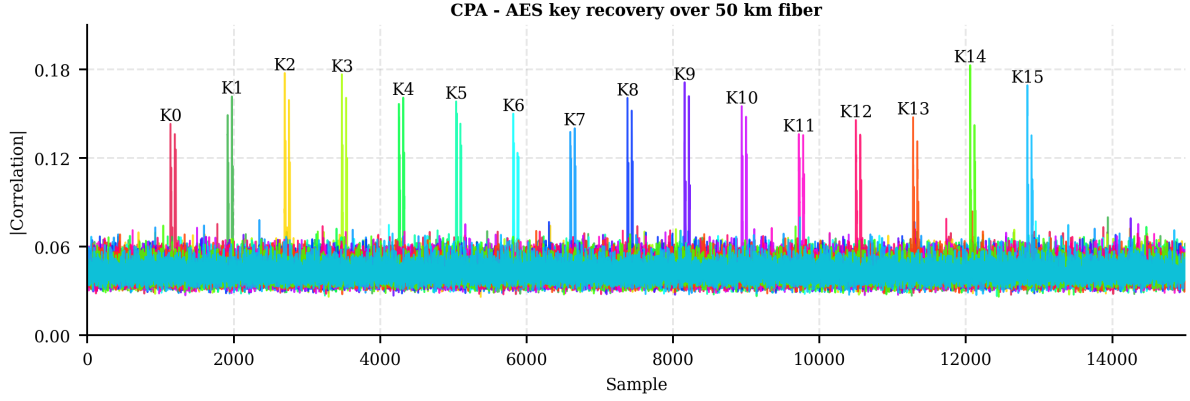


Fig. 2: Absolute Pearson correlation coefficient as a function of sample index for all 256 key hypotheses, shown for each of the 16 key bytes after 50 km propagation. On top of the spikes, K_n label represents the n -th key byte.

A 1577 nm optical signal at 5 dBm from a DFB laser is transmitted in a 50 km single mode fiber. The optical received power by a PIN photodiode, is -12.8 dBm resulting in a 17.8 dB optical link budget (2 dB loss of connectors included).

The photodiode output current is preamplified by a Radio Frequency (RF) preamplifier (30 dB gain) and then is measured by a digital Teledyne Lecroy oscilloscope (4 GHz bandwidth, 40 GS/s, 8-bit vertical resolution).

Results

A Fast Fourier Transform to the temporal traces, shows a peak at the fundamental clock frequency (16 MHz) of the Arduino (Fig. 3), indicating the existence of an optical side-channel carrying information of the microcontroller's activity.

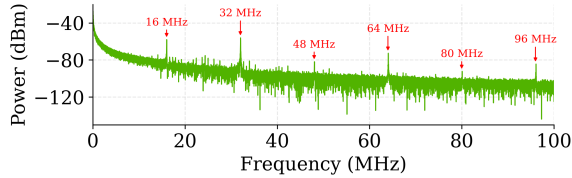


Fig. 3: Power spectrum of the mean optical trace after 50 km propagation (with RF preamplifier).

We apply a Correlation Power Analysis (CPA) on this leakage signal which is a divide-and-conquer attack that targets each of the 16 AES key bytes independently, testing all 256 possible hypotheses [7]. For the first SubBytes operation, the model value is based on the Hamming weight of the S-box output:

$$h_{i,k} = \text{HW}(S(p_i \oplus k)) \quad (1)$$

where i denotes the trace index, p_i is the plaintext byte value for the i -th trace, k is the candidate key byte (0 to 255), $S(\cdot)$ is the AES SubBytes substitution function, $\oplus$ is the bitwise XOR operation, and $\text{HW}(\cdot)$ is the Hamming weight.

For each of the 256 key hypotheses and each time sample, the Pearson correlation coefficient between the model values and the measured optical amplitudes is computed. A high correlation at a

given time sample indicates that the optical power fluctuation at that instant is statistically correlated to the cryptographic activity predicted by the model. Fig. 2 shows the absolute Pearson correlation for all 256 key hypotheses across the 16 key bytes with 5000 traces for clarity. The correct key byte reaches a correlation of 0.13 to 0.20, well above the noise floor of incorrect hypotheses (~ 0.03 to 0.07). This process is repeated for all 16 bytes, recovering the full 128-bit key.

To quantify the minimum number of traces required for reliable key recovery, we compute the first-order Success Rate (SR) [8]: the probability that the correct key byte hypothesis is ranked first among all 256 candidates, averaged over all 16 key bytes and repeated experiments (Fig. 4). The number of traces required for SR to reach 1.0 increases with fiber distance without any pre-amplifications: convergence is fast in the back-to-back configuration, slower at 12.5km, and even slower at 50km. Adding an RF preamplifier at 50 km reduces the number of traces required. We observed that the cryptographic activity remains detectable on the optical carrier, and that the limiting factor is the vertical resolution of the oscilloscope's 8-bit ADC, which struggles to accurately digitize the small power fluctuations at low received signal levels.

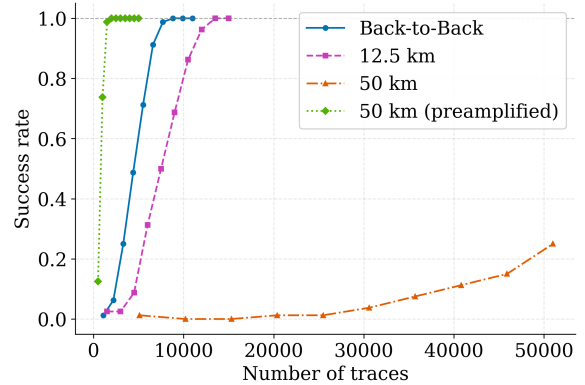


Fig. 4: Key Recovery Success Rate as a Function of Number of Traces for Different Fiber Distances

Discussion

As previously stated, traditional side-channel attacks require close physical proximity to the target device [9], the Screaming Channels attack reaches tens of meters over RF [5], and LED-based optical attacks require line-of-sight within visible range [4].

Here, we demonstrate that our approach relies on a transmission of the leakage signal into an optical fiber, extending the attack reach to 50 km which is a significant finding as it reveals a class of vulnerability that was previously considered impractical at such distances.

It is also worth noting that this attack is not mitigated by Quantum Key Distribution (QKD). While QKD secures the key exchange channel against interception, the leakage demonstrated here occurs during AES data encryption, after the key has been distributed. An attacker observing the fiber can recover the AES-128 key directly from the encryption activity, regardless of how that key exchange was established. This underscores that physical-layer side-channel hardening of the encryption hardware remains necessary even in QKD-secured optical links.

Conclusion

It has been demonstrated that AES-induced power fluctuations in a microcontroller, can modulate a co-powered DFB laser and can be detected after propagation through a 50 km single mode fiber. Using a first-order CPA on 1,500 optical traces, the full AES-128 key has been recovered.

To the best of our knowledge, this is the first long-range optical side-channel attack demonstrated at access-network distances, revealing that information leakage can propagate far beyond the local hardware environment. This attack has a high interest in the PON context, where the broadcast nature of the downstream signal means that any subscriber on the optical tree can easily and passively intercept the leakage signal, making this attack a fully passive and undetectable eavesdropping scenario. Although this proof of concept uses a microcontroller as a surrogate for the encryption hardware, the power coupling mechanism between the encryption logic and the laser transmitter is physically plausible in commercial OLT modules, where both components are typically powered from the same supply.

In future work, we plan to generalize this proof of concept in the context of PON networks. Extending this attack to commercial OLT hardware will require addressing trigger-free synchronization, which could be achieved by exploiting the PON frame structure, and adapting the analysis to hardware-accelerated AES implementations running at higher clock frequencies.

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